

GREATER BINGHAMTON, NY, USA

WMO: 725150

Lat: 42.207N

Lon: 75.980W

Elev: 486

StdP: 95.62

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 04725

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.1	-15.6	-22.9	0.5	-16.9	-20.3	0.7	-14.6	11.0	-2.6	10.0	-5.1	4.5	300	0.546

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.4	29.5	21.1	28.0	20.3	26.6	19.7	22.6	27.1	21.8	25.8	21.0	24.8	4.2	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.2	16.8	24.7	20.4	16.0	23.9	19.7	15.3	23.2	68.9	27.1	65.9	25.8	62.9	24.8	26.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.2	8.3	7.5	-20.9	32.0	3.1	1.5	-23.2	33.0	-25.0	33.9	-26.7	34.8	-28.9	35.9		
			-21.3	24.4	3.1	1.2	-23.5	25.3	-25.3	26.0	-27.0	26.6	-29.3	27.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.3	-5.2	-3.9	0.4	7.3	13.6	18.3	20.7	19.8	16.0	9.7	3.5	-1.9
	DBStd	10.32	6.50	5.75	6.06	5.35	4.71	3.64	2.93	2.95	3.98	4.69	5.19	5.29
	HDD10.0	1936	473	389	304	114	17	0	0	0	3	64	203	369
	HDD18.3	3908	730	622	556	334	160	46	11	18	93	268	444	627
	CDD10.0	1299	1	1	7	32	129	248	331	304	181	56	8	1
	CDD18.3	231	0	0	0	2	14	44	83	64	21	2	0	0
	CDH23.3	1425	0	0	2	23	110	277	522	362	125	5	0	0
	CDH26.7	290	0	0	0	4	18	54	117	73	23	0	0	0
Wind	WSAvg	3.7	4.2	4.3	4.3	4.2	3.6	3.3	3.0	2.9	3.2	3.7	3.9	4.1
Precipitation	PrecAvg	963	63	58	74	85	88	100	87	84	92	76	82	74
	PrecMax	1245	162	114	152	218	179	240	195	172	245	201	191	155
	PrecMin	769	19	13	18	18	20	25	37	36	16	7	26	24
	PrecStd	140	31	27	30	41	42	54	43	34	50	46	35	33
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.1	13.8	20.9	26.5	28.8	30.6	31.5	31.2	29.4	23.7	18.3	15.4
		MCWB	12.8	10.1	13.6	15.7	19.3	22.0	22.2	21.4	20.1	17.7	13.1	12.7
	2%	DB	10.1	9.7	15.8	22.2	26.4	28.2	29.4	28.7	26.6	21.2	15.7	11.2
		MCWB	8.3	5.8	9.9	13.5	17.7	20.5	21.4	20.8	19.6	15.9	12.1	9.1
	5%	DB	6.9	7.0	12.6	18.8	24.0	26.5	27.9	26.8	24.5	19.2	13.8	8.2
		MCWB	5.3	4.1	8.4	11.7	16.8	19.9	20.7	19.9	18.7	15.2	10.6	6.2
	10%	DB	3.6	4.2	9.6	16.0	21.6	24.7	26.3	25.3	22.4	17.2	11.7	5.5
		MCWB	2.2	1.9	5.8	10.2	15.4	18.7	20.0	19.4	17.7	13.8	9.2	3.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.8	11.2	14.1	17.1	21.1	23.2	24.1	23.6	22.4	19.6	15.0	13.0
		MCDB	13.9	13.6	19.4	23.9	27.1	28.0	29.1	28.1	26.4	22.0	16.7	15.1
	2%	WB	8.9	6.8	11.3	14.9	19.5	22.0	22.7	22.2	20.9	17.7	13.0	9.6
		MCDB	9.7	9.0	14.6	19.8	24.3	26.4	27.5	25.8	24.8	19.6	15.0	11.0
	5%	WB	5.4	4.4	8.8	13.1	17.9	21.0	21.9	21.3	19.8	15.8	11.3	6.7
		MCDB	6.8	6.5	12.2	17.7	22.1	24.9	26.0	24.9	22.9	18.5	13.2	8.0
	10%	WB	2.3	1.9	6.1	10.9	16.6	19.9	21.1	20.4	18.7	14.0	9.2	3.8
		MCDB	3.5	3.9	9.5	14.5	20.6	23.4	24.8	23.9	21.3	16.8	11.4	5.4
Mean Daily Temperature Range	5% DB	MDBR	7.1	7.7	8.5	9.9	10.1	9.5	9.4	9.3	9.5	8.3	7.2	6.1
		MCDBR	9.4	10.6	12.3	13.9	12.5	11.2	10.7	10.7	11.3	10.6	10.0	8.7
		MCWBR	8.8	8.5	8.1	7.8	6.5	5.6	4.9	4.8	5.4	6.4	7.3	7.6
	5% WB	MCDBR	9.2	9.8	11.6	12.7	11.2	9.9	9.4	9.1	9.6	9.2	9.1	8.5
		MCWBR	8.9	8.5	8.6	8.4	6.6	5.6	5.0	4.8	5.2	6.5	7.7	8.2
Clear-Sky Solar Irradiance	taub	0.289	0.306	0.328	0.350	0.381	0.431	0.437	0.420	0.356	0.343	0.302	0.284	
	taud	2.385	2.339	2.390	2.391	2.334	2.215	2.207	2.245	2.411	2.452	2.522	2.479	
	Ebn at Noon	857	897	918	919	893	845	835	839	876	845	842	838	
	Edh at Noon	80	98	106	114	124	140	140	131	103	88	70	67	
All-Sky Solar Radiation	RadAvg	1.48	2.35	3.44	4.40	5.20	5.52	5.71	5.05	4.07	2.59	1.70	1.18	
	RadStd	0.12	0.32	0.31	0.51	0.52	0.52	0.42	0.29	0.43	0.30	0.21	0.15	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (28 neighbors)	+0.39	N/A	N/A	N/A	N/A	N/A	N/A	-104	+105	+46

Nomenclature: See separate page